

5.1.5 Components of Deep Learning

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنے...

CS XII

Chapter 5: Application and Impacts of Computing

Lecture 06 of 15

5.1.5 Components of Deep Learning

LEARN IT

Deep learning relies on specific advanced components to achieve its high accuracy. These components **differentiate it from basic machine learning**.

#	Component	What it means
1	Large input data	Deep learning is " data hungry ." It needs massive datasets (thousands or millions of records) for better learning and accuracy.
2	Many hidden layers	Uses dozens or hundreds of hidden layers to learn deeper, more intricate patterns.
3	Automatic feature learning	Unlike older models where humans had to manually highlight important features, deep learning extracts important features by itself .
4	Learning algorithm	Math routines (like backpropagation) that help the system reduce mistakes and improve learning continuously.
5	High computing power	Processing millions of data points across deep layers needs specialised, strong hardware, such as GPUs (Graphics Processing Units).

6	Training process	The model is trained again and again in loops called epochs to gradually improve its final performance.
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A **GPU** was originally made for video games' graphics, but it can do thousands of small multiplications in parallel — exactly what deep networks need. An **epoch** is one full pass through the training data. Ten epochs means the model has seen every training example ten times, adjusting weights a little each time.

Backpropagation means sending the error (from the loss function) **backwards** through the layers so each weight can be nudged in the direction that reduces the next error.

WORKED EXAMPLE

Worked example: cotton-leaf disease detector for a farm in Mirpurkhas. Match each DL component.

1. **Large input data.** Researchers collect 80,000 labelled photos: healthy leaves and leaves with whitefly damage.
2. **Many hidden layers.** The network has 20 hidden layers so it can go from "green edge" to "this exact pest pattern."
3. **Automatic feature learning.** No agronomist types in "look for yellow circles." The network finds the useful spots itself.
4. **Learning algorithm.** After a wrong guess ("healthy" when it was diseased), backpropagation lowers the weights that caused the mistake.
5. **High computing power.** Training runs on a GPU workstation at a university lab in Karachi; a school laptop would take weeks.
6. **Training process.** They train for 30 epochs. After epoch 1 accuracy is 62%; after epoch 30 it is 94% on photos it has not trained on.

REMEMBER IT

D-L-A-L-G-T. Data (lots), Layers (many), Automatic features, Learning algorithm, GPU power, Training epochs. "Deep Learners Always Love Good Training."

COMMON MISTAKE

Mistake: "Deep learning is just 'a neural network'."

Fix: Name the six extras: huge data, many hidden layers, automatic features, a learning algorithm (backpropagation), GPUs, and repeated epochs. A shallow NN with a small spreadsheet is ordinary ML, not DL.

ڈیپ لرننگ کو عام مشین لرننگ سے یہ چیزیں الگ کرتی ہیں: بہت بڑا ڈیٹا، بہت سی پوشیدہ تہیں، خود بخود خصوصیات نکالنا، سیکھنے کا الگورتھم جیسے بیک پروپیگیشن (Backpropagation)، طاقتور ہارڈویئر جیسا کہ جی پی یو (GPU)، اور بار بار تربیت جسے ایپوک (Epoch) کہتے ہیں۔

CHECK YOURSELF

1. Why is deep learning called "data hungry"?
2. What is an epoch?
3. Name the hardware often used for DL, and the learning algorithm named in the source.

Answers: 1. It needs massive datasets (thousands or millions of records) for accuracy. 2. One training loop — one full pass through the training data. 3. GPUs (Graphics Processing Units); backpropagation.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. DL components include:

- a. Deep stacks of layers, lots of data, lots of compute, specialised architectures
- b. A single if
- c. A single Access form
- d. A single NAND

MCQ 2. CNNs are often for:

- a. Images
- b. Undo stacks
- c. Printer queues
- d. Primary keys

MCQ 3. RNNs/transformers often for:

- a. Sequences (text/speech)
- b. Pie charts
- c. ER diamonds
- d. K-map groups

MCQ 4. GPUs help because:

- a. Many multiply-adds in parallel
- b. They store FKs
- c. They draw OSI
- d. They boil water

MCQ 5. Pretrained models let you:

- a. Reuse features (transfer) when you have less data
- b. Skip ethics
- c. Skip labels forever
- d. Skip HCI

MCQ 6. A practical school limit is:

- a. Not enough labelled images/GPU
- b. Too many GPUs always
- c. Too few ethics issues
- d. Too much labelled data always

Extra short questions

1. Name DL-specific components vs a tiny NN.
2. Why not train ImageNet in a college lab from scratch?
3. One architecture name at definition level.
4. Ethics still apply.

Extra long question

1. A team wants DL for Urdu speech marks entry. List DL components they need and three reasons to start smaller (ML or rules).

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Depth, specialised layers (conv), big datasets, accelerators, regularisation.
2. Compute and labelled data; use a pretrained net if anything.
3. CNN for images / transformer for language — no need for full maths.
4. Medical DL that is biased still harms patients.

Long

1. Need: audio data, labels, model, GPU, evaluation. Start smaller: n small, accents, privacy of children's voices, electricity/cost. Human checks the marks.

Next lecture: 5.2 Applications of Neural Networks and Deep Learning